SECTION – 19

TECHNICAL SPECIFICATION OF LINE WORKS

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SECTION – 7
TECHNICAL SPECIFICATION OF LINE WORKS

PART-I : GENERAL:

1.1 **SCOPE :**

This part of the specification provides for check survey, distribution of all the material to work locations, setting to work, erection, testing, commissioning and for performance guarantee of EHV Transmission line as mentioned in the summary sheet.

1.2 The length of the line and the quantities specified in the Schedule-B against various items may vary depending upon the site conditions that may be obtained actually and also the changes / deviations in the alignment of the line that may have to be carried out depending upon the right of way, forest clearance, PTCC Clearance and other problems that may arise in the course of execution of the contract. The contractor shall execute the works to the extent required and shall have to complete the work as a whole (the Transmission line which is the ultimate work to be put into use). For any deviations in quantities arising due to changes in alignment or other reasons, the contractor will not be eligible for anything extra over the bided / accepted rates.

2.0 **ROUTE :**

The actual route of the transmission line shall depend on the results of the survey being carried out by the APTRANSCO and is also subject to the approval of the Power Telecommunication Coordination Committee. For any deviation in the alignment for any reason whatsoever the bidders are not eligible for any extra rate over the bided rates.

3.0 **TOPOGRAPHY:**

Along the route of the transmission line the altitude of the country varies from sea level to about 1000 meters above mean sea level. The topography of the terrain is fairly plain having hilly terrain, cultivated fields, gardens, and small thick jungles.

4.0 **TYPE OF CONSTRUCTION :**

The line shall be constructed on self-supporting galvanised lattice steel towers suitable for assembly at site by bolt connections. The conductors shall be in vertical formation. Continuous earth wire shall be provided above the conductors for effective shielding etc. The construction details shall comply in all respects with the specification given herein.

The line shall initially terminate on terminal towers situated at a distance of not more than 75 meters from the out-door substations at either end. From the terminal towers, the line shall be connected to the terminal structures in the sub-stations.

5.0 **PERMITS AND PRIORITIES:**

The contractor shall make his own arrangements in using private roads, pathways, etc., in connection with the construction work. The contractor shall himself arrange for permits required for the operation of the vehicles used in construction works. The APTRANSCO may, however assist the contractor in obtaining controlled commodities necessary for the execution of the works if it considers the demand justified. No extension shall however, be admissible to the contractor in the completion schedule for the erection of the transmission line, if such permits or priorities are not granted.

6.0 CEMENT AND STEEL REINFORCEMENT RODS:

The cement and steel reinforcement rods required for the work have to be supplied by the contractor as per latest ISS. Ordinary Portland cement of Grade-43 manufactured as per IS-8112 of 1989 shall only be used. Storage, certification, delivery and testing of the cement shall confirm to IS-8112 with latest amendments. IS Certification mark is obligatory. Cement procured from major cement manufacturing plant such as M/s.Raasi, M/s.Vishnu, M/s.Andhra Cement Company, M/s.ACC, M/s.CCI, M/s.L&T, M/s.Coromandal, M/s.Priyadarshini, M/s.Madras Cements etc will only be accepted for which prior approval shall be obtained from the APTRANSCO. Reinforcement Steel shall conform to IS 1786 and shall be procured from standard manufacturers i.e. TISCO, SAIL & VSP for use in tower foundations. **Copies of invoices and test certificates from the cement and steel manufacturers shall be submitted by the contractor to the Engineer and obtain clearance before actual use**. Such clearance will be given within a maximum period of one week.

7.0 REPLACEMENT :

In the event of the materials lying with the contractor for the execution of this contract being lost, damaged or destroyed while being in the custody of the contractor before being taken over by the APTRANSCO, the contractor shall be liable to make good the loss without any extra charges to the APTRANSCO.

8.0 COMPLETENESS OF BID:

The bidder should submit the bid complete in all respects. No handling, idle, or storage charges shall be paid by the APTRANSCO to the contractor. The bidders should include these in their quoted rates for individual items and / or the works.

9.0 SCHEDULES AND APPENDICES:

The details of the approximate quantities of material supplies and erection works involved are given in the Schedules A & B respectively. The bidders are required to quote for the quantities mentioned in the Schedules. However, the extent of material supplies as well as the erection works shall be decided based on the actual field conditions and therefore, for unit rates for each item shall form the basis of contract for payment purpose.

10.0 SUPPLEMENTAL ITEMS AND PROCEDURE FOR WORKING OUT RATES FOR SUPPLEMENTAL ITEMS.

Refer Clause (18) Section- I, General in Volume-I

11.0 SPECIAL WORK :

The rates for works not included in the schedule will be decided upon, when any such necessity arises during the execution of the works, by negotiations between the Engineer and the contractor and the contractor shall perform the work on the terms and conditions contained in this specification. The bidder must inform himself of the conditions of the terrain / area and quote the rates.

12.0 DATE OF COMPLETION :

It is of utmost importance that the line shall be completed as per the provisions in Schedule – D of this specification.

If the line is not completed by the contractor within the time stipulated in the specification duly adhering to the programme as given by him, damages / penalty for the delay in completion will be recovered as provided in the specification.

13.0 MATERIALS REQUIRED FOR THE WORK :

This is a total turnkey contract and the Successful bidder shall supply all the material required for erection of the line. The material is to be delivered by the contractor at the site stores. The contractor will be responsible for the proper handling and maintenance of the materials up to the time of handing over of the completed works to the APTRANSCO and return of surplus materials if any, at the APTRANSCO's stores. The cost of such handling and storage shall be covered in the quoted prices detailed in Schedule-A.

After completion of the Check Survey, the contractor has to furnish the actual quantity of material required for each item covered in Schedule-A and proceed to procure the requisite quantity of material only. Should however, on completion of all the works, any surplus materials for which payments have been made by the APTRANSCO are lying with the contractor, all such material shall be handed over to the Engineer at the APTRANSCO's stores. This shall be done within one to two months after completion and handing over of the line. If the materials are not returned within this period and / or in the opinion of the Executive Engineer in charge of the work, are not in a fit condition for use, such surplus materials will be treated as sold to the contractor at the rate equal to the stores issue rate plus 10% centage charge or at the prevailing market rates plus incidental charges, whichever is higher, and recoveries made accordingly. The 10% retention amount against the contract will be released only after all the balance materials are handed over at APTRANSCO's stores in proper shape and material accounts are rendered.

14.0 INSURANCE FOR THEFT AT CONTRACTOR'S STORES /TRANSIT LOSSES ETC.

It is the entire responsibility of the contractor to ensure safety for the materials lying with him for erection purpose. The contractor has to insure the materials during erection at his cost and the contractor will be responsible for the safety of the materials till the line is handed over for operation.

15.0 ERECTION TOOLS :

The erection tools required for the construction of the line shall be arranged by the contractor at his own cost. The bidder shall furnish full details of the tools & plant available with him for carrying out the foundation works (details of form boxes available for concreting), tower erection works and stringing works in the form 'schedule-VII' enclosed to this specification.

16.0 ERECTION LOSSES :

The following maximum additional quantities of materials are permissible, to cover damages and losses during erection.

i) Disc insulators required : Up to 1% of the quantity.

ii) Conductor and earthwire : Up to 1% of the route length of the line.
(The 1% wastage includes the additional lengths required to account for sag and jumpers).

iii) Bolts and Nuts, washers and Hardware & accessories required for conductor & earthwire. : Up to 2% of the quantity

17.0 SURPLUS MATERIALS :

On completion of the works all materials left surplus from those supplied shall be handed over to APTRANSCO at the APTRANSCO's stores. If any surplus towers are to be devoluted, they must be in full shape or else cost of full tower will be recovered from the contractor. The contractor shall furnish a statement giving full details of all materials utilized on works (based on tower schedule which should give location wise details with types of towers, spans, foundations etc.) and the balance to be returned to the APTRANSCO and obtain a clearance certificate from the Executive Engineer, within one or two months depending upon the volume of work involved. It may be noted that returning of all materials and rendering of accounts for the materials received is a pre-condition for release of payment against final bill and also the 10% retention amount at a later date.

18.0 DEVIATIONS FROM SPECIFICATION:

Should a bidder desire to depart in any respect from the provisions of this specification, he must specifically mention such departures in his bid under the heading "BIDDER'S COMMENTS ON AND DEPARTURES FROM THE SPECIFICATION AND CONDITIONS OF CONTRACT". The bidder must explain in detail each and every departure he proposes to make. All departures shall be subject to the approval of the Engineer, otherwise the provisions of this specification shall prevail.

19.0 RESPONSIBILITY OF THE CONTRACTOR:

The contractor shall guarantee and be entirely responsible for the execution of the contract in accordance with the general conditions of contract, specification, schedules and appendices. He shall further guarantee and be responsible for proper erection within the guaranteed completion and maintenance periods.

20.0 PERFORMANCE GUARANTEE:

The contractor shall guarantee the satisfactory performance of the erected line for a period of 12 months from the date of handing over and the formal acceptance of the completed line by the APTRANSCO. If any defects are found within this period due to fault in erection, the contractor shall rectify such defects forthwith at site without any extra cost to the APTRANSCO.

The performance of the jumpers on the angle towers will be watched during the period of performance guarantee and for the failure of a jumper during this period resulting in the breakdown of the line an amount of Rs.5,000/-per jumper will be recovered from the bills of the contractor due for payment by the APTRANSCO against any of the works executed by him for the APTRANSCO.

The performance of suspension insulators string at suspension towers will also be watched during the performance guarantee period and in case it is found during the stage of operation of line that there is a fall of a suspension insulator string along with the hanger during the performance guarantee period an amount of Rs.2,000/- for such failure will be recovered from the contractor from any of his bills.

21.0 STANDARD SPECIFICATIONS:

Unless otherwise specified, works shall be carried out in accordance with relevant Indian standards and A.P.S.S. or any revision thereof which may be issued during the currency of the contract, and all the clauses relating to materials and workmanship, responsibilities, liabilities, commencement of works, completion delays and extensions, suspension works, and forfeitures etc., as provided in the A.P.S.S. shall be binding.

22.0 STANDARDS OF MATERIALS TO BE SUPPLIED BY THE BIDDER

The following shall be the standards of materials to be supplied by the bidder, which are covered under Schedule – B for erection of the lines.

Sl.No.	Indian standard Specification	Title
1.	IS-8112 of 1989 of Latest issue	Ordinary port land cement of grade 43
2.	IS-1786	Tor steel for reinforcement
3.	IS-456	For concrete aggregates.
4.	IS-383	For sand for use in rendering, plastering and morter
5.	IS-1239 (part-I)	Hot dip galvanized M.S.Pipe and galvanizing conforming to ISS-2633/1986
6.	IS-5613 (Part-II)	Danger Boards, Phase plates, Bird Guards and Number plates
7.	IS-3043	Pipe earthing
8.	IS-2062-1992	Steel for General structural purpose - For M.S.Flat 50 x 6mm for earthing of tower lines.

a) The material shall be procured from the regular suppliers of APTRANSCO
b) The material shall be procured after taking approval of SE/TLC/APTRANSCO
c) In respect of tower accessories, the contractor has to use the material on the line only after approval by SE/TLC.

PART-II : ERECTION

23.0 CHECK SURVEY:

The detailed surveys are carried out by the APTRANSCO. The successful bidder is required to carryout only check survey. After the approved profiles are received, the contractor must fix up locations based on the approved spans indicated in the profile / tower schedule and only conspicuous variations in the chainage and physical features as indicated in the profile and as actually noticed during check-survey must be brought to the notice of the APTRANSCO's Engineer for taking remedial action. The position of locations stacked on the ground should be got approved by the Executive Engineer concerned. If the site of a tower location is not suitable for locating a tower the location will have to be shifted suitably keeping in view the limitation of span and this will be decided by the Executive Engineer. All the tools and instruments required for carrying out the check-survey should be arranged by the Contractor himself.

24.0 SITE STORES:

The successful bidder shall establish temporary stores at his cost at a suitable location for storing the line material within 15 days of award of contract. This store should be dismantled and site cleared after the work is completed by him.

25.0 WAY LEAVES AND TREE CUTTING :

Payment of compensation for damaged crops or fruit bearing trees shall be initially made by the contractor, which will be reimbursed by APTRANSCO later on. The compensation so payable in respect of damaged crops shall be based on the assessment of MRO. In respect of the trees, compensation so payable is the lowest of the rates fixed by the Horticulture/ NABARD/ MRO. The contractor shall take up the payment of compensation amount only after written approval by the Executive Engineer. The payment shall be made in the presence of representatives of TLC, Accounts and Vigilance wings of APTRANSCO. The Superintending Engineer/TLC shall finalise the compensation claims.

The contractor shall take all possible steps to see that standing crops etc. are not damaged during execution of works. When such damage is inevitable the compensation will be borne by the APTRANSCO to the extent of land required for laying of foundation, provided the damage is with prior concurrence of the Executive Engineer. The contractor shall bear the compensation for damages caused without prior concurrence of the Executive Engineer concerned. Any extra land required by the contractor temporarily for dumping of the construction material and consequent damage to the crops etc. shall be the responsibility of the contractor.

In the event of any obstruction being encountered from the local villagers or authorities, the contractor shall immediately notify the Executive Engineer who shall take such steps as may be necessary to clear the obstruction. The contractor or his representative or gangs shall not adopt antagonistic attitude towards the village authorities with whom they may come into contact, but shall immediately report to the Executive Engineer any case of obstruction which cannot be settled amicably. No trees shall be cut until the APTRANSCO has made necessary arrangement with the authorities concerned and permission is given to the contractor to fell such trees. The APTRANSCO shall arrange to remove the obstacles as soon as possible. At times however, it may not be possible for the APTRANSCO to arrange right-of-way for laying foundations or erecting the towers or stringing the line. At all such times, the contractor shall shift his gangs to other locations. The rates quoted shall cover all such contingencies and no extra payments shall be claimed for such contingencies.

26.0 STUB SETTING AND FOUNDATIONS:

26.1 Setting of stubs

The contractor shall undertake full responsibility for the correct setting of stubs in accordance with the approved methods of the exact locations and alignment and in precisely correct level. Stub setting templates shall be used for proper setting of stubs.

Cement concrete foundations shall be constructed complying with the foundation drawings to be furnished to the successful bidder. The type of foundation to be laid at each location will be decided by the Executive Engineer after inspection of site and for this purpose the locations must be excavated for dry type foundations initially and offered for inspection. Foundation at a location shall be laid only on receipt of written communication from the Executive Engineer indicating the type of foundation to be laid.

In case foundations of types other than those mentioned in Appendix-I are found necessary in the course of execution of contract, such foundations shall have to be laid only on communication of drawings for such foundations by the Engineer.

26.2 The rate for laying the foundations with 1:2:4 mix concrete is inclusive of the cost of dewatering of pits during concreting but shall be exclusive of the excavation cost and inclusive of cement cost

The APTRANSCO reserves the right to uncover and examine any foundation if it is found to be not according to specification. The contractor shall be required to open up and rectify all defective foundations thereby discovered at his sole expense and cost.

26.3 Excavation for foundations:

Earth Work Excavation rates for foundation pits shall be quoted by the bidders for the following soils, including dewatering of pits. However rates for shoring and shuttering shall be quoted by the bidder separately against respective item of Schedule.

a) Dry soils.
b) Wet soils.
c) Partially Submerged soils.
d) Fully Submerged soils.

Hard Rock : Hard rock will be that which requires drilling and blasting by any method. The cost of drilling and blasting shall be included in the quoted rates. In certain cases, when the area around the location happens to be an inhabited area consequent to which blasting cannot be resorted to, the excavation may have to be done either by the process of drilling, wedging, hammering and splitting or by the process of heating and splitting by sudden cooling. The rate quoted shall take into consideration such contingencies also.

NOTE :
i) For soils other than hard-rock, excavation rate is inclusive of backfilling the pits with excavated soils after laying foundations and levelling of site.

ii) For soils covered by hard-rock variety, the excavation rate is inclusive of backfilling with excavated rock bits and borrowed earth duly rammed after laying the designed rock foundation.

iii) No extra charges shall be admissible for the removal of the fallen earth in the pit, when once excavated.

The estimated rates loaded with tender percentage should cover all contingencies during the process of excavation. For any reasons whatsoever, no extra claim will be accepted.

26.4 TYPE OF FOUNDATIONS:

Cement concrete footing shall be used for all types of towers in conformity with the present day practices followed in the country. There will be mostly pyramid, stepped and chimney type. In very loose soils and wet locations R.C.C type of foundations may have to be provided to suit actual conditions. All the four footings of towers will be similar. No separate rate is admissible for dewatering of pits for laying concrete. Bidders shall quote for concreting including cost of dewatering.

In case of river crossings it may be necessary to use well foundations or other special type of foundations based on actual conditions.

26.5 CLASSIFICATION OF FOUNDATIONS:

Classification of soil shall be made according to IS : 200 (Part I) 1974 for footing cast in open pits. The foundation designs for various types of Foundations as given below shall be adopted in the construction of all 132 kV & 220 kV Transmission Lines and shall be based on the following instruction. The Foundation Drawings as finalized by the Civil Engineering Wing for various soil conditions suitable for different Types of Towers being deployed in the construction shall be followed by all field officers strictly.

Sl.No.	**Soil at the Location of Tower**	**Classification of foundation to be adopted.**
1.	Where normal dry cohesive or non-cohesive soils are met.	Dry
2(a).	Where sub-soil water is met at 1.50 meters or more below the ground level in normal soil	Wet
(b)	Where surface water could stand for long period with water penetration not exceeding 1.0 m below ground level (e.g. Paddy fields, Sugarcane fields)	
3(a).	Where sub-soil water is encountered between 0.75 meter and 1.50 meters depth below ground level.	Partially Submerged / Partially Black Cotton
(b)	Where top layer of Black Cotton soil extends up to 50% of the depth with normal soil thereafter	
Note – 1	Where soil is clayey type, not necessarily black in colour, which shrinks when dry, swells when wet.	
4(a)	Where sub-soil water Table is within 0.75 m depth from ground level.	Fully Submerged – Black Cotton
(b)	Where top layer of Black Cotton soil exceeds 50% and extends up to full depth or is followed by normal soil.	
(c)	Where top layer is normal soil up to 50% of the depth but the lower layer is a Black Cotton soil	
Note – 2	In case of items 4(a), (b) & (c) above, the concerned Superintending Engineer (Civil) shall personally inspect and certify on soil classification and shall not delegate this responsibility to any sub-ordinate officers.	
5(a)	Where decomposed or fissured rock, hard gravel, kanker, limestone, laterite or any other soil of similar nature is met.	Fissured / soft rock
(b)	Where top layer of normal dry soil extends up to 85% of the depth followed by fissured rock without presence of water.	
6(a)	Where Hard rock is encountered at 1.5 m or less below ground level.	Hard Rock
(b)	Where Hard rock is encountered form 1.5 m to 2.5 m below Ground Level though Top layer is of normal soil	
(c)	Where chiseling, drilling and blasting is required for excavation.	

Note – 3	While classifying foundation of Wet, Partially Submerged / Partially Black Cotton, Fully Submerged / Black Cotton foundations mentioned above, the worst conditions may be considered and not necessarily the conditions prevailing at the time of inspection. For instance, there are areas where sub-soil water rises when canal water is let-out in the field raising sub-soil water to a considerable degree. Similarly the effect of monsoon or when the nearby reservoirs are full should be considered and not the conditions prevailing in open season or summer when work is carried out normally. In all such cases, the Chief Engineer (Civil) shall personally inspect and certify the requirement of higher classification on case to case basis.
Note – 4	For critical locations of cut-point / Angle Towers, the classification of Tower Foundations may be at next higher type for ensuring safety of the transmission line as likely to be subjected to variable forces from time to time on account of change in season and climatic conditions. This shall be decided by the concerned Superintending Engineer (Civil) on case to case basis on personal inspection.
Note – 5	In case of highly collapsible & loose soils and river crossings, special design for foundations are to be made by Civil Engineering Wing for adoption on case to case basis.
Note –6	At locations subjected to water logging, particularly in paddy fields, ponds, marshy areas, etc., the chimney of all stubs to be raised to 600 mm over the normal chimney being adopted in other cases. This shall be done after complete erection of towers as a separate item for identified locations.

In addition to the above, depending on the site conditions more varieties of foundations may be introduced suitable for intermediate conditions under the above classifications to effect more economy.

26.6 FOUNDATION CONCRETE:

(a) Pad Concrete: The cement concrete used for the base/pad of foundations shall be of 1:4:8 mix with 40mm well graded hard broken granite metal.

(b) Foundation Concrete: The cement concrete used for the foundations (frustrum/stepped and chimney) shall be of 1:2:4 mix ratio with 20 mm (3/4") hard broken granite metal.

The cement required for the work shall be supplied by the Contractor. The cement to be used for this line shall confirm the specification as per clause (20.1) of Section-5.

The sand used for the concrete shall be composed of hard silicon materials. It shall be clear and of a sharp angular grit type and free from earthy or organic matter and deleterious salts. The aggregate shall be of clean broken hard granite or other stone specified or approved by the Engineer. It shall be of hard, close grained quality. It shall also be as far as possible cube like, preferably angular, but not flaky, perfectly clean and free from earth organics or other deleterious matter. 20 mm aggregate shall be of size as will pass through 20 mm square mesh measured in clear.

The water used for mixing concrete shall be fresh, clean and free from oil, acids and alkali. Saltish or blackish water should not be used.

The concrete should be mixed as stiff as the requirements of placing the concrete in the forms or moulds with ease warrant and the degree to which the concrete resists segregation will permit. The quantity of water used should be the minimum.

In the interest of speedy execution of works and to obtain homogeneous mix of concrete, mechanical mixers and vibrators shall be used for mixing & consolidation of concrete, except in case of emergency (such as failure of mixer or where it is not practicable to have mixer up to the location) or for each concrete sub-base where hand mixing can be resorted.

Proper forms of moulds adequately braced to retain proper shape while concreting should be used for chimney or steps and slab portions. The mould should be made watertight so that cement cream will not come out leaving only sand and jelly consequently forming honeycombing in the concrete. The rate for concrete is inclusive of form box. Sufficient number of form boxes for each type of foundation should be made so that the works are not held up on this account.

The stub angle shall be free from rust and cleaned thoroughly and coated with cement paste made of one part of cement and 3/4 part of water (cement slurry) to a thickness of 1.6 mm (1/16") before the cement concrete is laid against the stub angles. The coating with cement slurry shall be done each time to such a height so that concrete will be laid to that height before the cement wash becomes dry. The form boxes shall be cleaned and oiled before these are used for concreting.

The concrete shall be laid in 150 mm layers and consolidated well so that the cement cream works up to the top and no honey-combing is left in the concrete. After concreting the chimney portion to the required height, the top surface should be finished smooth, with slight slope towards the outer edge to drain off the rainwater falling on the copings. These copings and muffings should be done after the stringing is completed in respect of tension locations and such other locations as may be decided by the Engineer at site. For such locations an adhoc reduction of Rs.100/- per location will be made while billing for concrete and this will be released after the coping and muffings are completed.

In special cases, the APTRANSCO Engineer at site may authorise use of quick setting or rapid hardening compound in certain foundations. The cost of such quick setting compound will be paid extra as per actuals (on production of valid vouchers) if the contractor procures and uses it on the works, subject to the certificate of the Executive Engineer at site regarding its actual consumption for the works. In case the APTRANSCO supplies the quick setting compound required, no extra cost will be paid to the contractor.

Concreting is to be done continuously and monolithically, so that the subsequent layers are laid before the initial setting of the bottom layer begins. If, on rare occasions, fresh concrete is to be laid on old concrete in less than a week, the surface of the old set concrete should be chipped and cleaned thoroughly with wire brush and washed with a layer of thick cement slurry before the new concrete is laid. If, however the concrete is more than 10 days old, the top layer of the set concrete should be chipped and cleaned thoroughly with wire brush and fresh water, and layers of cement mortar 12 mm thick shall be laid only after giving a coat of cement slurry, as specified above to ensure proper bonding between old and new concrete.

In wet locations, the site must be kept completely dewatered both during the placing of the concrete and for 24 hours after completion. There should be no disturbance of concrete by water during this period. No extra rate will be paid for the dewatering and the rate for concrete is inclusive of dewatering charges. The forms of moulds shall not be removed before a lapse of about 24 hours after the completion of concreting. After removal of the forms, the concreted surface, where required, shall be repaired with a rich cement and sand mortar in the shortest possible time.

26.7 CURING AND BACKFILLING:

The concrete after it is 24 hours old shall be cured by keeping the concrete wet continuously for a period of 14 days after laying it. No separate rate shall be payable for curing and it is to be included in the rate for concreting.

The pits may be backfilled with selected and excavated earth sprinkled with necessary amount of water and well consolidated in layers not exceeding 150 mm thickness after a minimum period of 24 hours on completion of concreting. As the uplift strength of the foundation depends upon proper consolidation of backfilled earth extreme care shall be taken in this regard with special emphasis. Thereafter both the backfilled earth and exposed chimney top shall be kept wet for the remainder of the prescribed time of 14 days. The backfilling of foundations shall be so carried out that after one monsoon the level of the backfilled earth does not go below the surrounding ground level. However if any foundation settles below ground level after one monsoon it would be filled up by the contractor to be in level with the surrounding ground without any extra charges to the APTRANSCO. The Stub setting templates shall be opened only after the completion of backfilling. No separate rate shall be payable for backfilling and it is to be included in the excavation rate.

After backfilling, about 150 mm high earthen embankment along the sides of excavation pits will be made and sufficient water will be poured in the backfilled pits so that standing water remains above the backfilled earth for at least 24 hours. The uncovered concrete chimney above the backfilled earth shall be kept wet continuously for 14 days after concreting by providing wet empty gunny bags fully wrapped around the concrete chimney for curing and ensuring that the bags are kept wet by frequent pouring of water on them. The excavated earth left over should be spread within the four legs of the tower location.

27.0 GROUNDING:

Pipe earthing or counterpoise earthing has to be done at such tower locations where it is found necessary depending upon the type of soil met with at the location and the earth resistance at that location.

27.1 PIPE EARTHING:

At locations where pits can be excavated the towers shall be earthed by pipe earthing as per the drawing enclosed to this specification. The grounding shall be effected by making about 325 mm dia and 3850 mm deep pit at a distance of not less than 3650 mm diagonally away from the stubs and filling in the pits with finely broken coke having granule sizes not more than 25 mm thick. The coke shall be maintained up to a distance of 150 mm from the pipe on all sides. The top edge of the pipe shall be at least 600 mm below the ground level. The steel strip shall be buried not less than 600 mm deep from the ground level. The earth flat shall be of G.I. as specified in the drawing enclosed.

The rate for earthing is inclusive of cost of GI earthing set, coke, salt, excavation, back filling and measuring tower footing resistance.

27.2 COUNTERPOISE EARTHING:

In places of high resistivity soils especially in rocky locations special earthing arrangement shall be employed in the form of counterpoise earth to bring down the tower footing resistance to 10 ohms. The counterpoise earth shall consist of 4 Nos. 7/8 SWG galvanised steel wires running radially from all the four tower legs for a distance of about 15 to 30 metres. The galvanised steel wire shall be connected to the tower legs by galvanized steel lugs forged or compressed at its one end, complete, with 16 mm (5/8") dia bolts and nuts, required for connecting the earthing to the tower and the counterpoise shall be buried radially from the tower base at 600 mm below/ground level. The lugs should be preferably buried in chimney portion of the foundation to avoid pilferage. The rate for counter poise earthing is inclusive of excavation, back filling, measuring tower footing resistance and materials such as lugs and exclusive of 120 mtr of steel wire.

27.3 MEASURING TOWER FOOTING RESISTANCE:

The tower footing resistance of all towers shall be measured by the contractor using his own instruments in dry weather after the erection in the presence of the Engineer before stringing of earth wire and furnish the same to this office.

28.0. ERECTION OF TOWERS:

Towers shall be erected by piecemeal method on the foundations, not less than 14 days after concreting and after such time that the concrete has acquired its full strength. The towers shall be erected in workmanship like manner and its members shall not be strained or bent during the course of erection. Care shall be taken to see that the jointing surfaces are clean and free from dirt or grit. The tower erection shall be done in strict accordance with the approved drawings.

After initial erection, all tower bolts shall be checked to ascertain that all nuts are fully tight. Suitable means shall be adopted by the contractor to ensure that none of the bolts have been left out. The bolt threads shall be punched immediately after completion of the erection of each tower to avoid the nuts becoming loose. Punching of bolts shall be made by chamfering the threads with center punch in at-least three places equally placed on the contact surface of bolts and nuts. All the bolts & nuts in the towers upto the height below the bottom cross arm level including the bolt connecting the cross bracings at the bottom cross arm level, shall be seam welded in the upper half round portion (25mm length) of the bolt & nut circumferentially by using 10 guage welding rods. This shall be done immediately after the tower is erected and its verticality is checked and found to be within the limits. After seam welding, one coat of zinc rich paint shall be applied on the seam-welded surface matching the galvanised portion of the tower.

The contractor shall entirely be responsible for the correct erection of all towers as per the approved drawings, and their correct setting on an alignment approved by the Engineer. If the stubs or superstructures after erection are found to differ from approved drawings or to be out of alignment, the contractor shall dismantle and re-erect them correctly at his own cost without extension of time. The towers must be truly vertical after erection and no straining will be permitted to bring them to verticality. Tolerance allowed for verticality is 25 mm in a height of 9.15 meters i.e., 1 in 360 of tower height. Verticality of each tower shall be measured by the contractor and furnished to the Executive Engineer.

Each tower shall be fitted with number plate, danger plate and phase plate. Anticlimbing devices shall be provided on towers erected at all railway crossings, important road crossings etc. Bird guards shall be provided on suspension towers. These accessories shall conform to the relevant standards.

29.0 INSULATOR HOISTING:

Suspension insulators shall be used upto deviations of 2 degrees on all tangent type towers in the line and tension insulators on all angle type towers.

For ensuring insulation co-ordination the suspension or tension strings on approach and terminal towers near the substation ends up to a length of 1.5 kilometers shall be provided with adjustable arcing horns on tower side and fixed arcing horns on line side, keeping the arcing gap as may be stipulated by the Engineer at site.

Insulator strings shall be assembled on the ground. These shall be cleaned and examined before hoisting. They shall be kept on tarpaulins at the tower site so that, once they are cleaned there is no scope for the insulator to get contaminated. All accessories shall be properly fitted. It shall be ensured that all current carrying parts are smooth, and without dirt, cracks, or chips. Those having glazing defects exceeding half-centimeter square shall not be used and such insulators shall be accounted for under 1% wastage allowed.

30.0 STRINGING OF CONDUCTOR AND EARTHWIRE:

The stringing of the conductors and earthwire shall be done by a standard method used for such lines, which shall be indicated in the bid. The bidders shall give details of the stringing method they propose to follow and indicate its adaptability and advantages. They shall also indicate the tools and equipment required for stringing by the method proposed by them. The contractor shall use his own stringing and erection tools and other equipments. Before the commencement of the stringing, the contractor shall submit the stringing charts for the conductor and earth wire showing the initial and final sags and tension for various temperatures and spans along with equivalent spans in the line, for approval by the Engineer.

The contractor shall be entirely responsible for any damage to the towers or the conductor during stringing. He shall also be responsible for proper distribution of the conductor drums to keep the numbers and lengths of cut pieces of the conductor to the minimum. Pieces leftover shall be accounted for under 1% wastage allowed.

30.1 HANDLING OF CONDUCTOR:

The contractor shall be entirely responsible for the proper handling of the conductor and earthwire drums in the field. Handling and transporting of the conductor drums shall be carried out in such a manner as to minimise the possibility of damage from abrasion through rough handling, or dirt and grit getting into reel and on the conductor, injury to conductor etc; particular care shall be taken at all times to ensure that the conductors do not get kinked, twisted or abrased in any manner. Drum lifting jacks shall be used for paving out conductor.

Trestles shall be provided in the spans in which the conductor is proposed for paving out and it shall be ensured that at no time the conductor touches the ground (whether hard or soft). If the conductor is damaged, the section affected shall be replaced or repaired by putting joint or using repair sleeve or polishing with emery cloth, so as to give satisfactory corona performance. At all stages of construction proper care shall be taken so that the conductor surface is smooth enough to give satisfactory corona and radio-interference performance.

All equipment used in handling and/or transporting the conductor, such as grips, pulleys, slings, cable car etc; shall be so designed and maintained that the surfaces which come into contact with the conductor, are kept free of foreign matters or defects which may lower the corona inception voltage of the conductor to an unacceptable level.

30.2 PULLING IN OPERATION:

The earthwire shall be strung and securely clamped to the towers before the conductors are drawn up, in the order of the top conductor first, then the middle and the bottom conductor at the end. It shall be ensured that all the conductors of one section should be identically tensioned. The pulling of the conductor into the travelers comprising aerial and ground rollers shall be carried out in such a manner that the conductor is not damaged or contaminated with any foreign substance and that it may not be rubbed with the rough ground surface. The travelers shall have a size and shape most suitable for the type of stringing and size of conductor involved and as recommended by the conductor manufacturer. These shall be approved by the Engineer before use. The travelers surface in contact with aluminum surface of conductor shall be of such a material that conductor is not damaged, like neoprene rubber, aluminum, or stainless steel. These shall be equipped with high quality ball or roller bearings for minimum friction. They should be inspected daily for free and easy movement in blocks during stringing and sagging.

During pulling out operation the tension in each conductor and earthwire shall not exceed the designed working tension of the conductor and the actual prevailing temperature. After being pulled, the conductor and earthwire shall not be allowed to hang in the stringing blocks for more than 96 hours before being pulled to the specified sag. During the time the conductors and earthwire are on the stringing blocks, before sagging in, it shall be ensured that the conductors and earthwire are not damaged due to wind, vibration or other causes.

30.3 SAGGING IN OPERATION:

The conductors and earthwire shall be sagged in accordance with approved stringing charts before they are finally attached to the towers through the earthwire clamps for the earthwire and insulator strings for the conductor.

The conductor shall be pulled up to desired sag and left in travelers for at least one hour after which the sag shall be rechecked and adjusted, if necessary, before transferring the conductors from the travelers to the suspension clamps. The conductors shall be clamped within 36 hours of sagging in. Prestressing of conductor is not required. Instead, the conductors should be strung to a sag of one inch to a foot less than the sag shown on the final sag charts provided by the erection contractor.

In respect of twin Moose conductors the adjustment of the sub-conductor sag by means of sag adjustment devices provided in the insulator string shall invariably be not permitted at the time of stringing of conductors. The sag adjustment plates shall be kept in fully open position at the time of stringing of conductor.

The sag will be checked in the first and last spans of the section in case of sections up to eight spans and in one intermediate span also for sections with more than eight spans. The contractor shall be responsible for any damage caused to the towers or to the conductors through overstressing during stringing. Dynamometers shall be used in checking the tension in the conductors and earthwire and these instruments shall be periodically checked with a standard dynamometer. The sag shall also be checked when the conductors have been drawn up and transferred from aerial rollers to the insulator clamps.

The stringing block, when suspended on the transmission structure for sagging, shall be so adjusted that the conductor on the traveler will be at the same height as the suspension clamp to which it is secured.

At sharp vertical angles, the sag and tension shall be checked on both sides of the angle. The conductor and earthwire shall be checked on the travelers for equality of tension on both sides. The suspension insulator assemblies shall normally assume vertical position when the conductor is clamped. Sagging operation shall not be carried out under wind, extremely low temperature, or other adverse weather conditions, which prevent satisfactory sagging.

The erection contractor should ensure that

i) Any departure from the correct sag in any span does not exceed 4 per cent.

ii) The sag of any conductor does not depart by more than 3 percent from the mean sag of all the conductors in the same span.

30.4 CONDUCTOR DAMAGE AND REPAIR:

If the conductor is damaged for whatever reason and the damage is not repaired by repair sleeves or by rubbing with emery cloth, it shall be brought to the notice of the Engineer and shall not be used without his approval. Even repairing of conductor surface shall be done only in case of minor damages, scuff marks etc., which are safe from both electrical and mechanical points of view. The final conductor surface shall be clean, and smooth, without any projections, sharp points, cuts or abrasions etc., for giving satisfactory corona and R.I performance.

All the joints on the conductor or the earthwire shall be of compression type, in accordance with the recommendations of the manufacturers for which the necessary tool and equipments like compressors and dies, grease guns, presses etc, shall have to be arranged by the contractor. These shall be made in the best workmanship like manner and shall be perfectly straight and having maximum strength. Each part of the joints shall be cleaned by wire brush to make it free of rust or dirt etc., and properly greased before the final compression is done with the compressors.

All joints or splices shall be made atleast 30 meters away from the structures. No joint or splice shall be made in spans crossing over main roads, railways, small rivers or in tension span. Not more than one joint per conductor shall be allowed in one span. The compression type fittings used shall be of self-setting type or care shall be taken to make the cable to indicate when the fitting is centered properly. During compression or splicing operation the conductor shall be handled in such a manner as to prevent lateral or vertical bearing against the dies. Care shall be taken to protect the conductor from scratches, abrasions or other damages. After pressing, the aluminum sleeve shall have all corners rounded, burrs and sharp edges removed and smoothened.

30.5 STRINGING RATE:

The stringing rate for stringing of the conductors per route length of line shall include paving, stringing, tensioning, clamping and jointing of conductors as per schedules and fitting of all necessary accessories, specified above or otherwise, which are normally required or are usual with such lines.

Similarly the rate for stringing of the earthwire shall also include the stringing and fixing of the necessary accessories like suspension and tension clamps, earth bonds, compression joints etc, as required. Wherever necessary ground clearance has to be measured to ensure obtaining adequate ground clearance and the rate should include this also. The performance of the jumpers on the angle towers will be watched during the period of performance guarantee and for the failure of a jumper during this period resulting in the breakdown of the line an amount of Rs.5,000/per jumper will be recovered from the bills of the contractor due for payment by the APTRANSCO against any of the works executed by him for the APTRANSCO.

30.6 ACCESSORIES:

Accessories like vibration dampers, armour rods etc., for the conductors shall also be fitted on the line as desired by the Engineer according to the recommendations of the manufacturers. Armour rods shall be provided at all suspension supports of the conductors and vibration dampers shall be provided on both sides of the suspension and tension supports at suitable distances from the supporting points for each phase conductor. While fixing these on the line care must be taken that conductor is not damaged in any way and no sharp edges are left at any point. All accessories shall be clean, smooth and in perfect condition before fitting.

30.7 FINAL CHECKING, TESTING AND COMMISSIONING:

After completion of the works final checking of the line shall be done by the contractor to ensure that all the foundation works, tower erection and stringing have been done as approved by the Engineer, and also that they are complete in all respects. All the works shall be thoroughly inspected, keeping in view the following main points.

I. Sufficient backfilled earth is lying over each foundation pit and it is adequately compacted.

II. Concrete chimneys and their coping are in good and finely shaped conditions.

III. All the tower members are correctly used strictly as per the drawings and are free from any defect or damage, whatsoever.

IV. All the bolts are fully tightened and they are properly punched and half-round welded as per contractual provisions.

V. The stringing of the conductors and earthwire has been done as per the approved sag and tension charts and desired clearances are available.

VI. No damage, minor or major to the conductor, earthwire, accessories and insulator strings still unattended are noticed.

VII. The jumpers provided at the cut points are connected rigidly to the tension hardware utilising all the jointing bolts provided for the purpose.
After final checking, the line shall be tested for insulation in accordance with tests prescribed by the Engineer. All arrangements for such testing or any other test desired by the Executive Engineer shall be done by the contractor and necessary labour, transport and equipment shall be provided by him. Any defects found out as a result of such test, shall be rectified by the contractor forthwith without extra charges to the APTRANSCO.

The contractor shall submit a report to the above effect to the Executive Engineer. In case it is noticed later that some or any of the above are not fulfilled, the APTRANSCO's Engineers will get such items rectified through other agencies and recover the cost of such works from the bills payable to the contractor against the contract or from retention amounts.
Recoveries will be affected in respect of the following also in addition to the above.

i)	For each bolt hole in a tower member which is not as per design	:	Rs.10/-
ii)	For each jumper failure at a cut point during the performance guarantee period.	:	Rs.5000/-

After satisfactory tests on the line and on approval by the Engineer the line shall be energized at full operating voltage before handing over.

30.8 RECTIFICATION OF DEFECTS DURING THE DEFECTS LIABILITY PERIOD:

After commissioning and handing over of the line to the APTRANSCO, the contractor shall arrange for thorough patrolling of the line during the first six months of the defects liability period so that the defects if any noticed during that period may be rectified by the contractor free of cost to the APTRANSCO without having to wait for the APTRANSCO to suggest such rectifications. During the balance period however, the contractor shall arrange for rectification of the defects free of cost to APTRANSCO as and when pointed out by the APTRANSCO.

30.9 EXECUTION OF WORKS RELATING TO POWER LINE CROSSINGS AND RAILWAY CROSSINGS:

Works such as erection of towers underneath an existing power line and paving out of conductors and earthwire and stringing the power line crossing span or a railway crossing span, will have to be done only after receipt of approval from the concerned officer, which, sometimes, may not match with the programme of the contractor.

In such cases, the contractor shall execute such works as and when approvals are received. His rates for tower erection and stringing shall take into consideration, such contingencies also.

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APPENDIX – A

I. TECHNICAL PARTICULARS OF 132KV LINE TOWERS

1.	Towers	Lattice type Galvanised DC towers(P R S type Kamani design).
2.	Normal span	320 Metres
3.	Number of Conductors	Three Conductors to be strung on ONE CIRCUIT
4.	Number of Earth wires	One
5.	Configuration of Conductors	Vertical formation
6.	Type of towers	i) Double circuit tangent towers up to 2 degrees deviation. ii) Double circuit angle towers up to 30 degrees deviation iii) Double circuit angle towers up to 60 degrees angle and dead-end towers.

II. Technical Particulars for Power Conductor:

Panther ACSR conductor

1.	Type	200 Sq.mm Copper equivalent '**Panther**' ACSR Conductor
2.	Stranding	30/3.00 mm Aluminum Over 7/3.00 mm Steel
3.	Overall diameter	21.00 mm
4.	Ultimate strength	9144 Kg
5.	Weight	976 Kg/Km

III. TECHNICAL PARTICULARS FOR GROUNDWIRE:

1.	Type	110 Ton capacity High Tensile galvanising steel wire
2.	Stranding	7/3.15 mm
3.	Overall diameter	9.45 mm
4.	Ultimate strength	5712 Kg
5.	Weight	428 Kg/Km

IV. Technical particulars for Insulators :

1.	Size of Disc	255 x 145 mm Ball and Socket type disc insulators for suspension strings and 280 x 145 mm for tension strings.
2.	Number of Discs	a) Suspension 13 Nos. b) Tension14 Nos.
3.	Electro Mech. Strength of insulators	a) Suspension strings 7000 Kgs. (or 70 KN) b) Tension strings 11500 Kgs. (or 120 KN)
	Use:	a) Suspension strings At all tangent towers. b) Tension strings At all angle towers.

WEIGHTS OF 132 KV TRANSMISSION LINE TOWERS and EXTENSIOS

Sl. No.	Structure Type	Approx Unit Weight in MT	Approx. No. of bolts & nuts upto bottom X-arm level
I)	**Type of Tower : P**		
1	Stub & Cleat for NT,+3,+6&+9	0.19	
2	SST for NT, +3,+6 & +9	0.44	
3	Normal Tower	2.81	276
4	+ 3 meters extensions	0.46	388
5	+ 6 meters extensions	0.97	504
6	+ 9 meters extensions	1.506	625
II)	**Type of Tower : R**		
1	Stub & Cleat for NT,+3,+6&+9	0.33	
2	SST for NT, +3 +6 & +9	0.94	
3	Normal Tower	4.36	440
4	+ 3 meters extensions	1.05	701
5	+ 6 meters extensions	2.32	960
6	+ 9 meters extensions	3.87	1230
III)	**Type of Tower : S**		
1	Stub & Cleat for NT,+3,+6	0.45	
2	SST for NT,+3,+6,	1.14	
4	Normal Tower	5.76	580
5	+ 3 meters extensions	1.35	811
6	+ 6 meters extensions	2.93	1041
7	+ 9 meters extensions	4.87	1342
IV)	**TNSA & TND/BNS/BNC Booms**		
1	TNS (including stub & cleat)	0.97	
2	TND (including stub & cleat)	1.15	
3	BNS / BNC	0.62	

I. TECHNICAL PARTICULARS OF TOWERS

1. Towers Lattice type Galvanised DC towers. (PRS type Kamani Design)

2. Normal span 320 Meters

3. Number of Conductors Three Conductors to be strung on ONE CIRCUIT

4. Number of Earth wires One

5. Configuration of Conductors Vertical formation

6. Type of towers
   i) Double circuit tangent towers up to 2 degrees deviation.
   ii) Double circuit angle towers up to 30 degrees angle towers
   iv) Double circuit angle towers up to 60 Deg. angle and dead-end towers.

II. Technical Particulars for Power Conductor:

		PANTHER ACSR	PANTHER AAAC
1. Type of Aluminium conductor	:	200 Sq.mm. Aluminium equivalent 'PANTHER' ACSR Conductor	241 Sq.mm. All alloy 'PANTHER'
2. Stranding Alluminium	:	30/3.00 mm Aluminium Over 7/3.00 mm Steel	37/2.88mm.
3. Overall diameter	:	21.00 mm	20.16 mm
4. Ultimate strength	:	9144 Kg	6749 Kg
5. Weight	:	976 Kg/Km	663.8 Kg/Km

III. TECHNICAL PARTICULARS FOR GROUNDWIRE:

1.	Type	110 Ton capacity High Tensile galvanizing steel wire
2.	Stranding	7/3.15 mm
3.	Overall diameter	9.45 mm
4.	Ultimate strength	5712 Kg
5.	Weight	428 Kg/Km

APPENDIX-IV

I. Different types of towers and their weights

Description	Appropriate weight per tower on MT	Approximate No. of bolts and nuts upto bottom cross arm level per tower
'P' Type tangent towers		
Normal	2.809	276
3 Mt. Extension	0.468	388
6 Mt. Extension	0.969	504
9 Mt. Extension	1.506	625
'R' Type **30⁰** angle towers		
Normal	4.357	440
3 Mt. Extension	1.048	701
6 Mt. Extension	2.319	960
9 Mt. Extension	3.865	1230
'S' Type **60⁰** angle deadend towers		
Normal	5.763	580
3 Mt. Extension	1.354	811
6 Mt. Extension	2.932	1041
9 Mt. Extension	4.871	1342

Contd....

II. Different types of foundations and their concrete volumes.

C. Description	Volume of Excavation (Cumt)	Volume of Concrete 1:4:8 (Cumt)	Volume of Sand filling (Cumt)	Volume of Concrete 1:2:4 (Cumt)	Reinforce-ment Steel (Kgs.)
For Tangent Towers (P – Type)					
Dry type	32.37	0.78	0	2.47	0
Wet type	59.25	1.60	0	5.76	0
Partially Submerged type	99.24	2.70	5.05	12.43	0
Fully Submerged type	132.43	3.72	6.73	15.26	202.34
Hard Rock type	2.56	0	0	2.80	0
Soft Rock type	15.55	0	0	4.91	0
For 30^0 angle towers (R – Type)					
Dry type	40.52	0.96	0	3.60	0
Wet type	92.83	2.5	0	8.80	0
Partially Submerged type	143.81	3.84	6.94	18.65	0
Fully Submerged type	179.63	4.90	8.66	21.62	336.29
Hard Rock type	4.34	0	0	5.76	0
Soft Rock type	35.04	0	0	8.85	0
For 60^0 angle towers (S – Type)					
Dry type	69.83	1.60	0	7.21	0
Wet type	135.17	3.36	0	18.14	0
Partially Submerged type	231.98	5.78	10.09	40.08	0
Fully Submerged type	261.13	6.56	11.35	49.13	487.82
Hard Rock type	9.86	0	0	11.13	0
Soft Rock type	51.20	0	0	18.78	0